

Basics 41: Phase Space and Distribution Functions

VI - Kinetic theory and energetic particles

From many-particle states to continuous kinetic descriptions, moments, invariants, and numerical marker representations

Textbook draft, version 1.0. Fluid variables describe averages, but energetic-particle resonance depends on where particles lie in both position and velocity. This chapter introduces six-dimensional phase space, defines the distribution function and its normalization, derives phase-space continuity and Liouville's theorem, and explains moments, reduced coordinates, Jacobians, and Monte Carlo marker weights.

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1 Learning objectives

After completing this note, the reader should be able to:

- Distinguish physical space, velocity space, and phase space.
- Interpret $f(\mathbf{x}, \mathbf{v}, t)$ as a number-density distribution.
- Recover density, flow velocity, pressure, and energy from moments.
- Derive phase-space continuity and Liouville's theorem.
- Transform a distribution between coordinate systems using the correct Jacobian.
- Explain full- f , marker-particle, and weighted Monte Carlo representations.

2 Prerequisites and reading strategy

Recommended prerequisites are Basics 1, 2, 10, 13, 19, and 22. Keep units explicit: in six-dimensional Cartesian coordinates, f has units of particles per physical-space volume per velocity-space volume.

3 Why phase space is needed

A point particle is specified by position and velocity,

$$\mathbf{z} = (\mathbf{x}, \mathbf{v}) \in \mathbb{R}^6. \quad (3.1)$$

Two particles at the same position can interact differently with a wave if their velocities differ. A fluid density $n(\mathbf{x}, t)$ therefore cannot represent resonant velocity selection, trapped and passing populations, pitch anisotropy, or energetic tails.

For N particles, the exact microscopic phase-space density is

$$f_{\text{micro}}(\mathbf{x}, \mathbf{v}, t) = \sum_{p=1}^N \delta(\mathbf{x} - \mathbf{x}_p(t)) \delta(\mathbf{v} - \mathbf{v}_p(t)). \quad (3.2)$$

Kinetic theory replaces this singular object by a smooth or coarse-grained distribution f .

4 Definition and normalization

The number of particles in a phase-space element is

$$dN = f(\mathbf{x}, \mathbf{v}, t) d^3x d^3v. \quad (4.1)$$

Therefore

$$N = \int f d^3x d^3v, \quad n(\mathbf{x}, t) = \int f d^3v. \quad (4.2)$$

If f is normalized as a probability density instead, the integral is one. A code and its documentation must state which normalization is used.

For a uniform isotropic Maxwellian,

$$f_M(v) = n \left(\frac{m}{2\pi k_B T} \right)^{3/2} \exp\left(-\frac{mv^2}{2k_B T}\right), \quad (4.3)$$

whose velocity integral equals n .

5 Moments

The mean of a phase-space function $A(\mathbf{v})$ at fixed position is

$$\langle A \rangle = \frac{1}{n} \int A(\mathbf{v}) f d^3v. \quad (5.1)$$

Important moments are

$$n = \int f d^3v, \quad (5.2)$$

$$\mathbf{u} = \frac{1}{n} \int \mathbf{v} f d^3v, \quad (5.3)$$

$$\mathbf{P} = m \int (\mathbf{v} - \mathbf{u})(\mathbf{v} - \mathbf{u}) f d^3v, \quad (5.4)$$

$$\mathcal{E} = \int \frac{mv^2}{2} f d^3v. \quad (5.5)$$

The pressure tensor contains more information than a scalar pressure. For a gyrotropic magnetized population,

$$\mathbf{P} = p_{\perp}(\mathbf{I} - \mathbf{bb}) + p_{\parallel}\mathbf{bb}. \quad (5.6)$$

6 Phase-space continuity

Let the phase-space velocity be

$$\dot{\mathbf{z}} = (\dot{\mathbf{x}}, \dot{\mathbf{v}}). \quad (6.1)$$

Conservation of particles in a phase-space control volume gives

$$\boxed{\frac{\partial f}{\partial t} + \nabla_{\mathbf{z}} \cdot (f\dot{\mathbf{z}}) = 0}. \quad (6.2)$$

Expanding,

$$\frac{Df}{Dt} + f \nabla_{\mathbf{z}} \cdot \dot{\mathbf{z}} = 0, \quad \frac{D}{Dt} = \partial_t + \dot{\mathbf{z}} \cdot \nabla_{\mathbf{z}}. \quad (6.3)$$

If the phase-space flow is incompressible,

$$\nabla_{\mathbf{z}} \cdot \dot{\mathbf{z}} = 0, \quad (6.4)$$

then

$$\boxed{\frac{Df}{Dt} = 0}. \quad (6.5)$$

This is the local form of Liouville's theorem: the distribution is constant along collisionless characteristics.

7 Liouville theorem for Lorentz motion

For nonrelativistic Lorentz dynamics,

$$\dot{\mathbf{x}} = \mathbf{v}, \quad \dot{\mathbf{v}} = \frac{q}{m}(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (7.1)$$

The six-dimensional divergence is

$$\nabla_{\mathbf{x}} \cdot \dot{\mathbf{x}} + \nabla_{\mathbf{v}} \cdot \dot{\mathbf{v}} = \nabla_{\mathbf{x}} \cdot \mathbf{v} + \frac{q}{m} \nabla_{\mathbf{v}} \cdot (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \quad (7.2)$$

$$= 0. \quad (7.3)$$

Thus collisionless Lorentz trajectories preserve phase-space volume.

8 Characteristics and filamentation

Even though f is constant along each trajectory, the distribution at a fixed point can change because characteristics move and shear. A smooth patch in phase space can stretch into fine filaments. This is phase mixing. Coarse numerical grids may erase those filaments and produce artificial entropy or damping.

9 Coordinate transformations and Jacobians

Suppose velocity coordinates change from (v_x, v_y, v_z) to spherical variables (v, ϑ, φ) . Then

$$d^3v = v^2 \sin \vartheta \, dv \, d\vartheta \, d\varphi. \quad (9.1)$$

For an isotropic distribution,

$$n = 4\pi \int_0^\infty f(v) v^2 \, dv. \quad (9.2)$$

In pitch coordinates $\xi = v_\parallel/v$,

$$d^3v = 2\pi v^2 \, dv \, d\xi. \quad (9.3)$$

A function that is constant in v is not a uniform probability density in speed because the shell volume grows as v^2 .

Guiding-center and flux coordinates have nontrivial Jacobians. The invariant number is always

$$dN = f_z(\mathbf{z}) d\mathbf{z} = f_y(\mathbf{y}) d\mathbf{y}, \quad (9.4)$$

so $f_y = f_z |\partial \mathbf{z} / \partial \mathbf{y}|$.

10 Reduced distributions

A reduced distribution is obtained by integrating over unresolved variables. For example,

$$F(\psi, E, \lambda) = \int f \delta(\psi - \psi(\mathbf{z})) \delta(E - E(\mathbf{z})) \delta(\lambda - \lambda(\mathbf{z})) \, d\mathbf{z}. \quad (10.1)$$

Reduced models are useful only if their measure and omitted dependencies are documented. A quantity called $f(E, \lambda)$ may represent a density with respect to $dE \, d\lambda$, not the original six-dimensional f .

11 Marker particles and weights

Monte Carlo methods represent a distribution with markers sampled from a convenient marker density $g(\mathbf{z})$:

$$f(\mathbf{z}) \approx \sum_{p=1}^{N_m} w_p \delta(\mathbf{z} - \mathbf{z}_p). \quad (11.1)$$

If markers are sampled from g , an observable is estimated as

$$\int A f d\mathbf{z} = \int A \frac{f}{g} g d\mathbf{z} \approx \frac{1}{N_m} \sum_{p=1}^{N_m} A(\mathbf{z}_p) \frac{f(\mathbf{z}_p)}{g(\mathbf{z}_p)}. \quad (11.2)$$

Importance sampling chooses g to place markers where the integrand matters, especially near resonances or loss boundaries.

12 Invariants and entropy

For collisionless incompressible phase flow,

$$\int C(f) d\mathbf{z} \quad (12.1)$$

is conserved for broad classes of functions C . Particle number and fine-grained entropy are examples. Coarse graining and collisions can increase observable entropy even though exact Vlasov evolution is reversible.

13 Connection to the Kinetic MHD-PINN project

Module 2 defines energetic-particle distributions and their normalization. Module 3 samples orbits. Module 5 evaluates resonant phase-space integrals. Module 6 evolves reduced distributions. Every interface must specify coordinates, Jacobians, units, and whether values represent pointwise f , cell-integrated particle number, or marker weights.

14 Computational laboratory

1. Sample a three-dimensional Maxwellian and verify density and temperature moments.
2. Evolve a one-dimensional free-streaming distribution $f(x, v, t) = f_0(x - vt, v)$.
3. Visualize phase-space filamentation for a sinusoidally modulated initial density.
4. Compare direct quadrature and Monte Carlo moment estimates.

5. Demonstrate the error caused by forgetting the v^2 Jacobian in an isotropic speed integral.

15 Summary

Phase space records both position and velocity. The distribution function is a density with respect to a specified phase-space measure. Collisionless Hamiltonian or Lorentz dynamics preserves phase-space volume, so f is constant along characteristics even while it develops fine structure. Fluid variables are moments of f , and numerical marker methods approximate those moments through weighted samples.

16 Exercises

1. Determine the SI units of a six-dimensional number-density distribution.
2. Verify the normalization of the Maxwellian.
3. Derive $d^3v = 2\pi v^2 dv d\xi$.
4. Show directly that Lorentz phase-space flow is incompressible.
5. Explain the difference between fine-grained and coarse-grained entropy.

Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
f	Distribution function	Particle number density with respect to a stated phase-space measure.
$\mathbf{z}$	Phase-space coordinate	Combined position and velocity coordinate.
$\mathbf{P}$	Pressure tensor	Second central velocity moment multiplied by mass.
g	Marker density	Probability density used to sample numerical markers.
w	Marker weight	Ratio or coefficient relating the represented distribution to marker density.

17 References

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2. N. A. Krall and A. W. Trivelpiece, *Principles of Plasma Physics*.
3. R. B. White, *The Theory of Toroidally Confined Plasmas*.
4. C. K. Birdsall and A. B. Langdon, *Plasma Physics via Computer Simulation*.